TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Driven Acoustic Sensing and Forecasting for Early Detection of Crop and Forest Pest Insects			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Software Solutions	Measuring Instruments	Measurement Sensors and Components
Concept of Proposed Technology	Pest monitoring in Korean forests and orchards is of critical strategic importance for biodiversity and economic factors (1), (2). Traditional monitoring of insects that damage crops and forests rely heavily on pheromone traps, manual counting, and static risk maps derived mainly from climate or historical data (3). These approaches are, on one hand, labor-intensive and often deadly for insects' population, and on the other hand, react late to local outbreaks and are difficult to scale across many farms and forest stands. In addition, conventional machine learning models deployed in the field often degrade over time and across sites because they are not designed to continuously adapt to new environments, new pest species, and management practices. To address these limitations, this project adopts an AI-driven acoustic sensing and forecasting platform for early detection of pest insects in crops and forests. Low-cost, autonomous acoustic sensor nodes will capture airborne and structure-borne sounds produced by insects (e.g., wingbeats, stridulation, boring and feeding activity), while edge AI models perform on-device			

	detection and classification. These local detections are combined with meteorological and publicly available remote-sensing data in spatio-temporal forecasting models that provide short-term risk maps and outbreak predictions. By focusing on cost-efficient acoustic sensing and robust AI adaptation, the system will be tested at scale in representative Korean environments, such as orchards, plantations, and forest stands. The resulting system can be integrated at a later stage into pest monitoring solutions of Korean SMEs operating in smart agriculture, forestry, and environmental IoT markets.
Rationale for Joint R&D Project	AI-based acoustic pest detection and outbreak forecasting must work reliably in very heterogeneous conditions: conifer and broadleaf forests, fruit orchards, and potentially greenhouses and rice-paddy landscapes. Soundscapes, pest species composition, and farming practices differ significantly between Europe and Korea, and even among the same landscape. Validating algorithms in only one country or a single type of environment is not sufficient to guarantee robust performance and generalization. The proposed international collaboration combines Fraunhofer IDMT's recent endeavors in pest monitoring (4) and expertise in acoustic sensing, signal processing, and AI with a Korean SME's practical knowledge of local forestry and agricultural operations and their product portfolios (e.g., smart traps, monitoring platforms, farm/forest management systems). Joint R&D enables systematic evaluation on real orchards and forest sites in Korea, with a particular focus on pine wilt–related beetles in pine forests and key pests in fruit orchards as relevant use cases (1), (2). The resulting technology can become a differentiating module in future Korean monitoring solutions and can be extended to other crops, forest types, and export markets in subsequent projects
Key Words	1. Acoustic-based Pest Detection 2. Edge AI and Field Adaptation 3. Spatio-Temporal Outbreak Forecasting 4. Smart Monitoring for Crops and Forests

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 6
R&D Objectives	The objective of this project is to develop an integrated AI-driven acoustic sensing and forecasting system for early detection of crop and forest pest insects that can be reliably deployed and maintained across diverse agricultural and forest environments. Specifically, the project aims to: *Objective 1.* Design robust and data-efficient acoustic detection algorithms that identify and classify key pest insects and activity stages (e.g., boring, feeding, flight) with minimal labeled data and limited communication bandwidth *Objective 2.* Develop spatio-temporal AI models that combine sensor detections with weather and other contextual data to produce short-term outbreak forecasts and risk maps at farm- and stand-scale *Objective 3.* Integrate detection and forecasting into an operational, low-touch platform that minimizes manual intervention, supports continual adaptation to new sites and seasons, and reaches a maturity level where prototypes can be produced and evaluated at a pilot scale, preparing the technology for subsequent integration into Korean SME products and services.			
R&D Target Specifications	The target performance specifications of this project are as follows: 1. Reduce manual counting and inspection effort per trap or monitoring point by at least **80%** compared to conventional visual inspection workflows. 2. Achieve at least a **15% increase in F1-score** for early detection of target pest insects in unseen sites relative to non-adaptive baseline models using the same hardware. 3. Reduce false alarm rates (false-positive outbreak alerts) by at least **40%**, improving trust and operational usability for farmers and forest managers. 4. Provide **short-, medium- and long-range outbreak risk forecasts** with calibrated probabilistic outputs without need for manual model retraining.			
R&D Plan	**Phase 1: Core Algorithms for Acoustic Pest Detection and Efficient Adaptation (TRL 3 → 4)** In Phase 1, the project focuses on developing core algorithms for acoustic detection of pest insects and rapid model adaptation to new crops, forests, and sound environments. The central question is how to exploit shared acoustic structure (e.g., between similar beetle or moth species) while adapting efficiently to site-specific factors such as background noise, sensor placement, and local species composition. The project will investigate few-shot and weakly supervised learning methods that use only a small number of labeled examples from new sites, along with parameter-efficient tuning strategies (e.g., feature adapters, lightweight fine-tuning) to update models without full retraining. Confidence-aware filtering and semi-supervised updating will be explored to leverage large volumes of unlabeled audio. Laboratory and controlled-environment experiments will be conducted to characterize typical acoustic signatures of relevant insect groups and to define optimal sensor configurations for edge deployment. The main milestone of this phase is the demonstration of stable and reproducible insect detection and classification performance under simulated field conditions and public benchmark datasets, showing that the proposed methods can significantly reduce labeled data requirements and computational cost while maintaining accuracy. **Phase 2: Spatio-Temporal Outbreak Forecasting and Continual Learning (TRL 4 → 5)**			

	While Phase 1 emphasizes local detection at the sensor level, Phase 2 extends the system to provide outbreak forecasts and risk maps for operational decision-making. The project will combine streams of acoustic detections with meteorological data and publicly available remote-sensing indicators (e.g., drought indices, vegetation condition) to build spatio-temporal AI models that predict short-term changes in pest pressure. Methods for mitigating memory loss effects and handling concept drift will be developed so that models can adapt to changing weather patterns, crop rotations, and pest population dynamics without losing past knowledge. This includes regularization-based continual learning, drift-aware updating triggered by statistical change detection, and replay-lite strategies using compact summaries of historical states. Uncertainty estimation and reliability monitoring will be integrated so that the system can flag low-confidence predictions and prioritize data collection where it is most informative. The milestone of this phase is the implementation of a forecasting framework that produces calibrated risk maps and outbreak predictions while maintaining stable performance over time and across regions, based on realistic Korean and European use cases. **Phase 3: Operational Platform and Field Validation with Korean SME (TRL 5 → 6)** Building upon the detection and forecasting capabilities developed in Phases 1 and 2, Phase 3 aims to integrate these technologies into an operational platform and validate them in real-world agricultural and forest environments in collaboration with the Korean SME partner. In this phase, the acoustic detection modules and spatio-temporal forecasting models will be integrated into a unified pipeline that automates data ingestion, quality control, drift detection, model adaptation, and the delivery of alerts and risk maps via APIs and dashboards. Edge device firmware and backend services will be aligned with the Korean SME's hardware and cloud infrastructure to support integration into their monitoring products in later stages. Low-touch deployment procedures and configuration tools will be developed to enable technicians and end users to install and operate the system with minimal AI expertise. Field-level validation will be conducted at multiple pilot sites provided by the Korean SME partner (e.g., pine forests affected by pine wilt–related beetles and fruit orchards with economically important insect pests). The system will be evaluated using practical KPIs such as reduction in manual inspection effort, early warning lead time, false alarm rates, user workload, and robustness under different weather and noise conditions. The final milestone is the demonstration of a scalable, cost-efficient, and AI-enabled pest monitoring and forecasting system at a pilot scale that could be commercialized by the Korean SME and replicated across diverse agricultural and forest environments in follow-up activities.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute partner (Fraunhofer IDMT) will lead the core algorithmic research, acoustic sensing concepts, and system-level design of the proposed technology. Its primary role is to develop and validate the AI methods that enable reliable acoustic detection of pest insects and spatio-temporal forecasting of outbreaks under real-world conditions. Specifically, the institute will: 1. Design and implement algorithms for acoustic signal acquisition and preprocessing, including sensor selection, noise-robust feature extraction, and on-device processing strategies. 2. Develop data-efficient and adaptive detection models capable of identifying target pest insects and activity patterns across different environments with limited labeled data. 3. Build spatio-temporal AI models that combine sensor detections with weather and contextual data to generate outbreak forecasts, risk maps, and uncertainty estimates. 4. Establish a unified operational platform architecture that integrates edge devices, data pipelines, model management, monitoring, and interfaces suitable for integration into Korean SME products in subsequent stages. 5. Conduct systematic benchmarking, ablation studies, and long-term evaluations to ensure scientific soundness, robustness under domain shifts, and compatibility with regulatory and data-privacy requirements. In addition, Fraunhofer IDMT will provide research infrastructure, computing resources, and expertise in acoustic signal processing, embedded AI, and trustworthy machine learning, and will be responsible for documenting the technology and supporting knowledge transfer to the Korean SME partner.
Expected R&R of Korean SME Partner	The Korean SME partner will play a central role in real-world deployment, field validation, user-driven requirements, and commercialization-oriented planning for the proposed technology. While the research institute focuses on core AI and system design, the SME ensures that the solution matches Korean agricultural and forestry practices and can be integrated into market-ready products in follow-up stages. Specifically, the SME partner will: 1. Provide representative test sites (e.g., orchards, plantations, forest stands, greenhouses) and real-world acoustic data to support realistic evaluation of detection and forecasting methods. 2. Lead the field installation and operation of sensor nodes and gateways, including power supply, communication, and integration into their existing monitoring platforms. 3. Define practical performance requirements and KPIs, such as acceptable false alarm rates, response times, preferred alert formats, and integration needs with farm/forest management workflows. 4. Co-design user interfaces and dashboards that present detections and risk maps in a clear, actionable way for Korean farmers, forest managers, and service providers. 5. Contribute to preliminary commercialization planning, including concepts for hardware and service packaging, pricing models, and after-sales support, and prepare internal roadmaps for future market validation in Korea and potentially other Asian markets. Through these roles, the SME partner will ensure that the technology is not only scientifically advanced but also aligned with the needs and constraints of Korean SMEs, thereby supporting rapid uptake and sustainable business models.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to generate both scientific and industrial impact. For AI and signal-processing researchers, the project will contribute to new methods for robust acoustic pest detection, data-efficient adaptation across sites, and spatio-temporal forecasting under sparse and noisy conditions. The combination of edge AI, continual learning, and outbreak modeling will advance the state of the art in adaptive environmental monitoring systems. For SMEs and technology providers in Korea, the developed solution offers a directly usable module that can be embedded into smart traps, IoT sensor networks, or existing monitoring platforms for crops and forests. By automating insect detection and providing actionable risk forecasts, the technology will help reduce manual labor, enable more targeted and timely interventions, and support new data-driven services such as subscription-based monitoring or advisory offerings. For end users such as farmers, growers, and forest managers, the system will provide earlier and more reliable warnings of damaging insect activity, helping to protect yields, reduce avoidable pesticide applications, and improve the resilience of forests and agricultural systems under climate stress. At a broader economic level, the international collaboration will strengthen the competitiveness of Korean SMEs in the field of AI-enabled environmental monitoring, foster technology transfer between Europe and Korea, and support the development of high-value, data-centric services in agriculture and forestry.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization